

Conclusion and Recommendations – Amazon Sales 2025

Conclusion

After performing Exploratory Data Analysis (EDA) on the Amazon Sales 2025 dataset, several important business insights were identified from the sales, customer, product, and transaction data.

1. Dataset Quality

- The dataset contains 250 records and 11 columns.
- No missing values were found in the dataset.
- No duplicate records were identified.
- The dataset is clean and suitable for analysis and visualization.

2. Product and Category Insights

- The dataset includes products from categories such as Electronics, Clothing, Footwear, Books, and Home Appliances.
- Electronics and Home Appliances generated the highest sales revenue compared to other categories.
- High-value products like Refrigerators, Laptops, Smartphones, and Washing Machines contributed significantly to total revenue.
- Low-cost products such as Books and T-Shirts had higher purchase frequency but lower revenue contribution.

3. Sales Performance Analysis

- The average total sales value is relatively high, indicating strong contribution from premium products.
- Maximum single-order sales reached 6000, showing that high-ticket items play a major role in revenue generation.
- Product quantity purchased ranges between 1 to 5 units, indicating moderate purchasing behavior.

4. Customer and Location Insights

- Customers are spread across multiple cities such as New York, Los Angeles, Chicago, Houston, Seattle, and others.

- Some locations contribute more frequently to sales, indicating strong regional demand.
- Repeat purchases of electronics and appliance products suggest strong customer preference for technology-related items.

5. Payment Method Analysis

- Multiple payment methods are used including Credit Card, Debit Card, PayPal, Amazon Pay, and Gift Cards.
- Digital payment methods are widely preferred, showing customer comfort with online transactions.

6. Visualization Insights

The visualizations created during EDA helped identify:

- Top-selling products
- Revenue contribution by category
- Product-wise quantity sold
- Distribution of total sales
- Customer purchasing patterns
- Product price comparisons

Overall, the business is highly influenced by premium electronic and appliance products, while low-cost products help maintain consistent order volume.

Recommendations

1. Focus More on High-Revenue Products

- Increase inventory and promotions for products like Laptops, Smartphones, Refrigerators, and Washing Machines.
- These products generate the highest revenue and should be prioritized for business growth.

2. Improve Marketing for Low-Performing Categories

- Categories such as Books and Clothing can be improved through:
 - Discounts
 - Combo offers
 - Festival sales
 - Personalized recommendations

3. Regional Sales Strategy

- Identify top-performing customer locations and run targeted advertising campaigns in those cities.
- Expand delivery and promotional services in high-demand regions.

4. Encourage Bulk Purchases

- Since quantity purchases are generally between 1–5 units, businesses can introduce:
 - Buy-more-save-more offers
 - Bundle pricing
 - Membership discounts

5. Enhance Digital Payment Offers

- Since customers prefer digital payment methods, provide:
 - Cashback offers
 - Wallet discounts
 - EMI options
 - Credit card partnerships

6. Customer Retention Strategy

- Build loyalty programs for repeat customers.
- Send personalized product recommendations based on previous purchases.
- Use email and app notifications for new launches and discounts.

7. Inventory Management

- Maintain higher stock levels for fast-selling and high-revenue products.
- Reduce overstocking of low-demand products to minimize storage costs.

8. Future Analysis Suggestions

Further analysis can improve business understanding, such as:

- Monthly and seasonal sales trends
- Customer segmentation
- Profit analysis
- Return/refund analysis
- Predictive sales forecasting
- Customer lifetime value analysis

